Computational Complexity Comparison via Scaling Limits for Emerging MCMC Samplers

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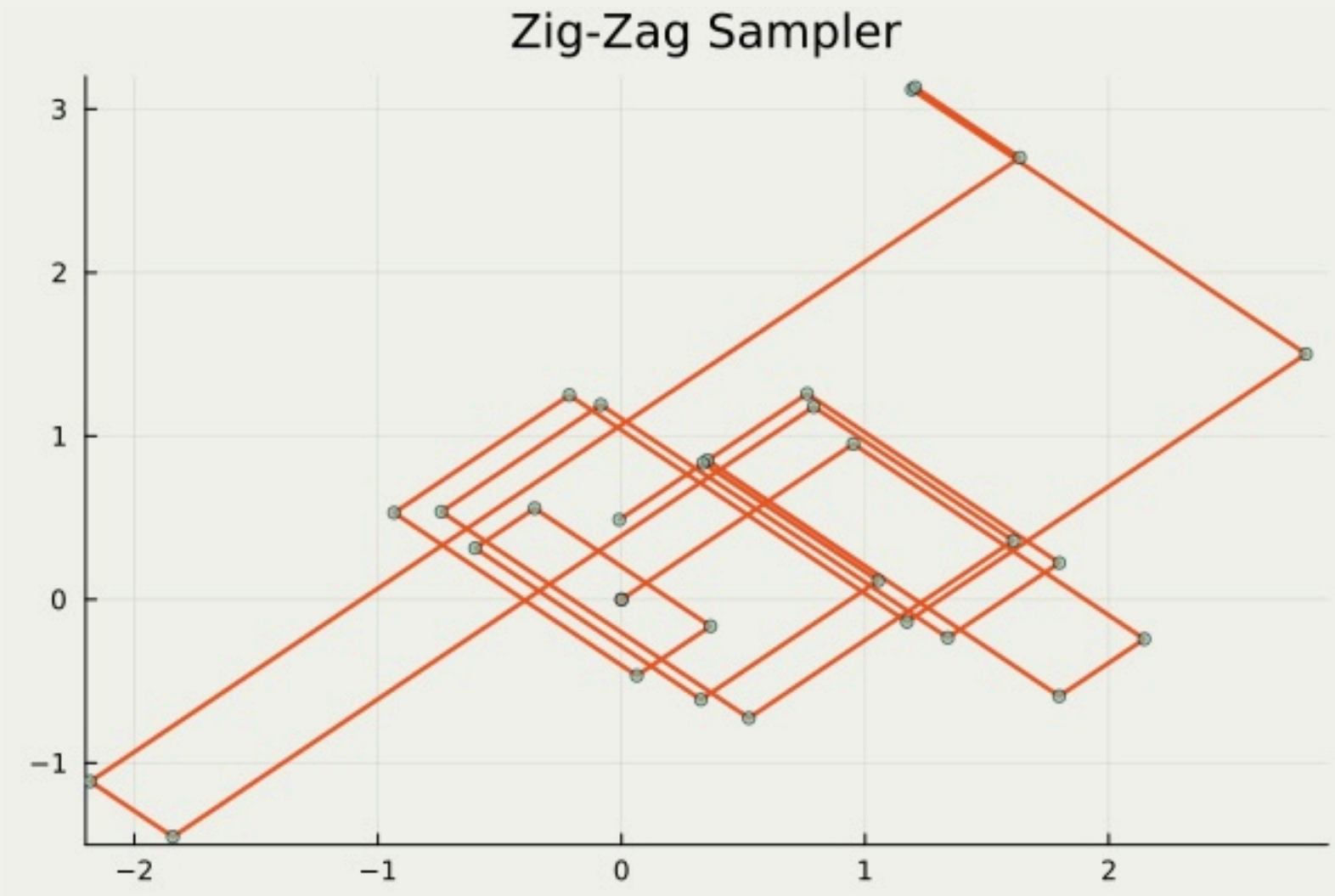
2/10/2026

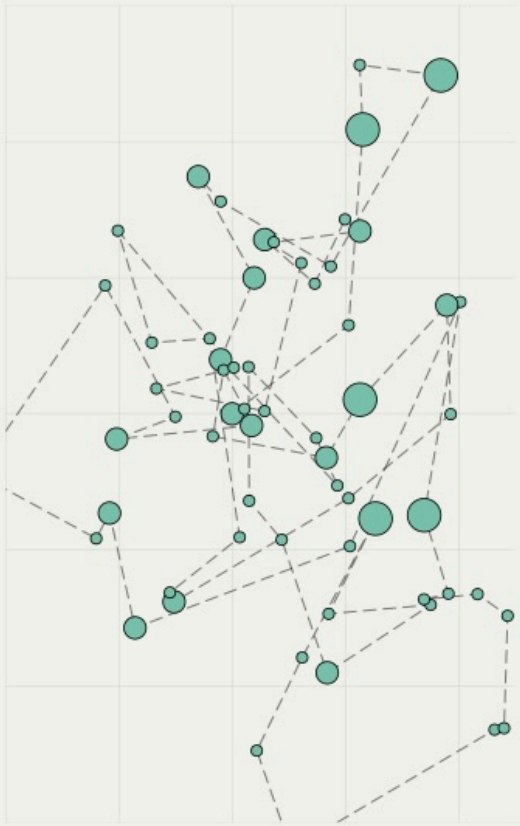
1 What is a PDMP?



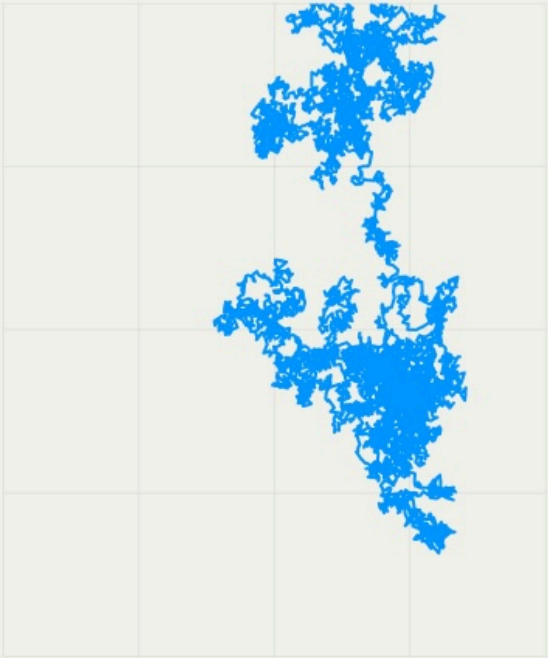
An illustration of a Piecewise Deterministic Markov Process. Output from the `PDMPFlux.jl` package.

1.1 Evolution of Monte Carlo

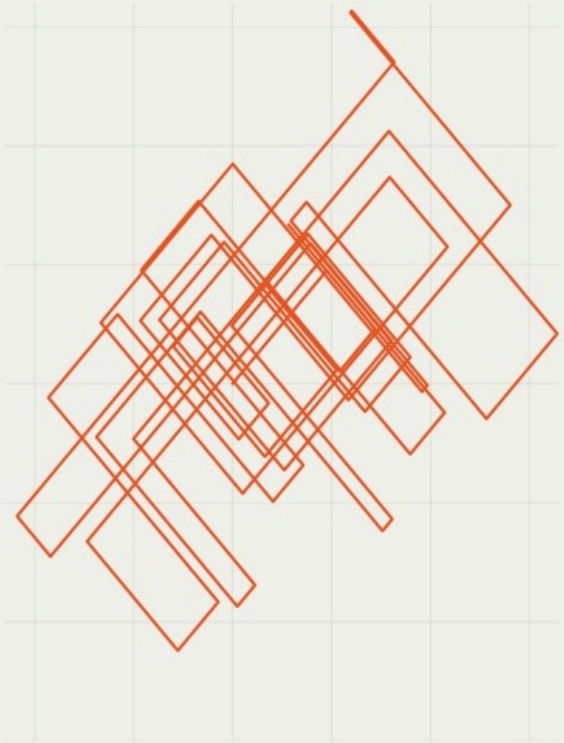
Random Walk Metropolis



Langevin Diffusion



Zig-Zag Sampler



Markov Chain
(1953–)

Diffusion
(1978–)

PDMP
(2008–)

1.2 PDMP ≒ ODE + Jump

$$X_n = X_{n-1} + \varepsilon_n \qquad n \in \{0, 1, 2, \cdots\}$$

Markov Chain = Discrete Time

Difference Equation

Markov Process = Continuous Time

Differential Equation

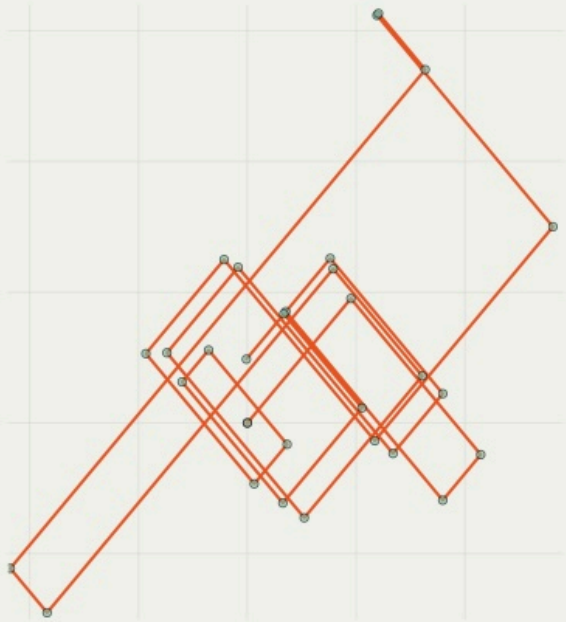
Drift **Diffusion** **Jump** $t \in [0, \infty)$

Diffusion $dX_t = a(X_t)\,dt + b(X_t)\,dB_t$

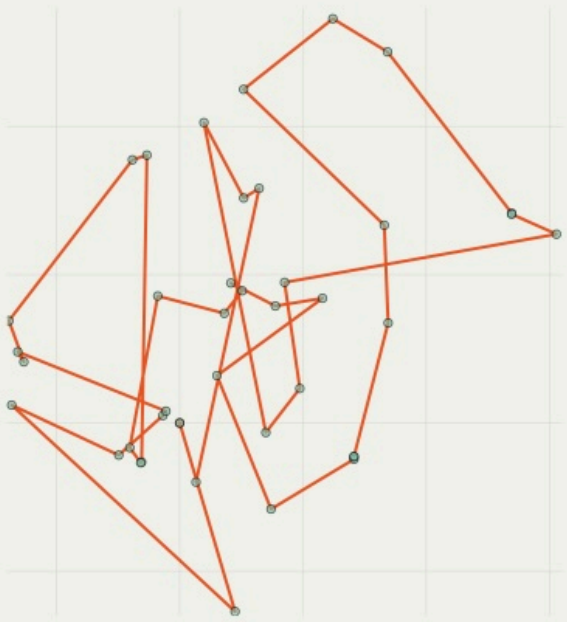
PDMP $dX_t = a(X_t)\,dt + \int_{\mathbb{R}^d} c(X_{t-}, u)\eta(dtdu)$

1.3 Which One is Faster ?

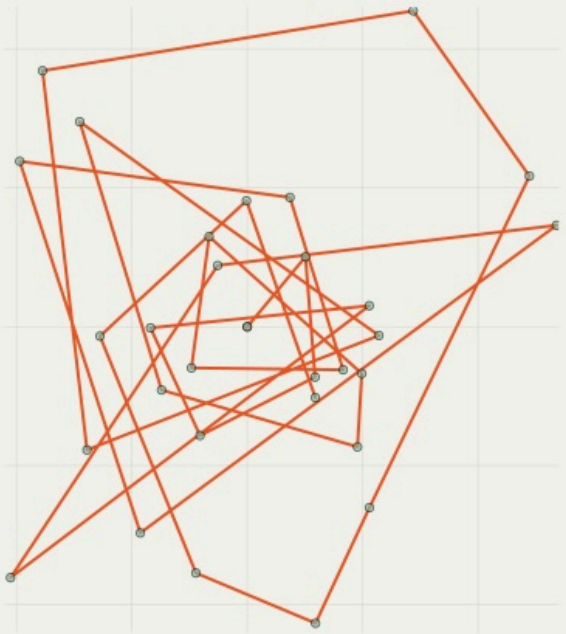
Zig-Zag Sampler



Bouncy Particle Sampler



Forward ECMC Sampler

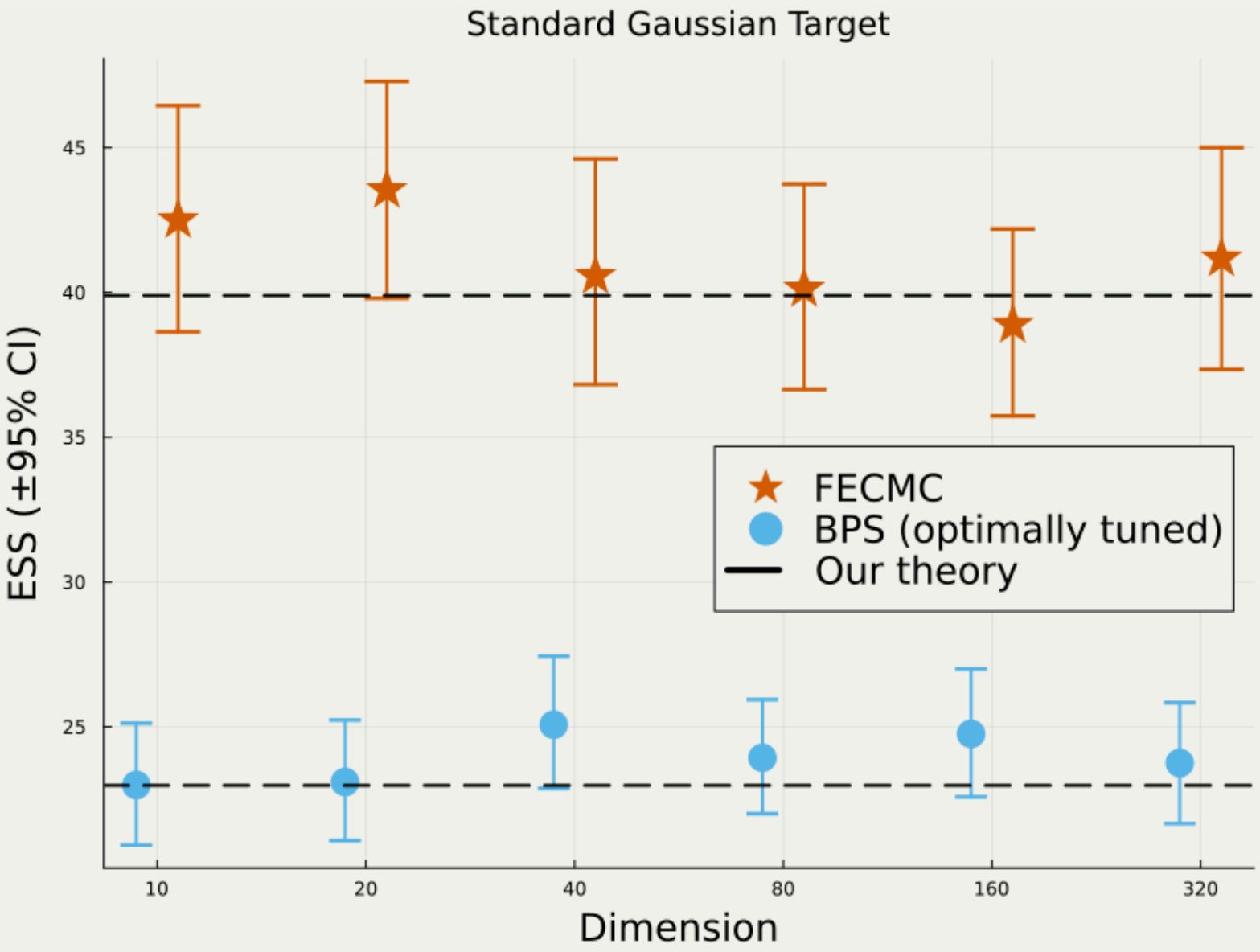
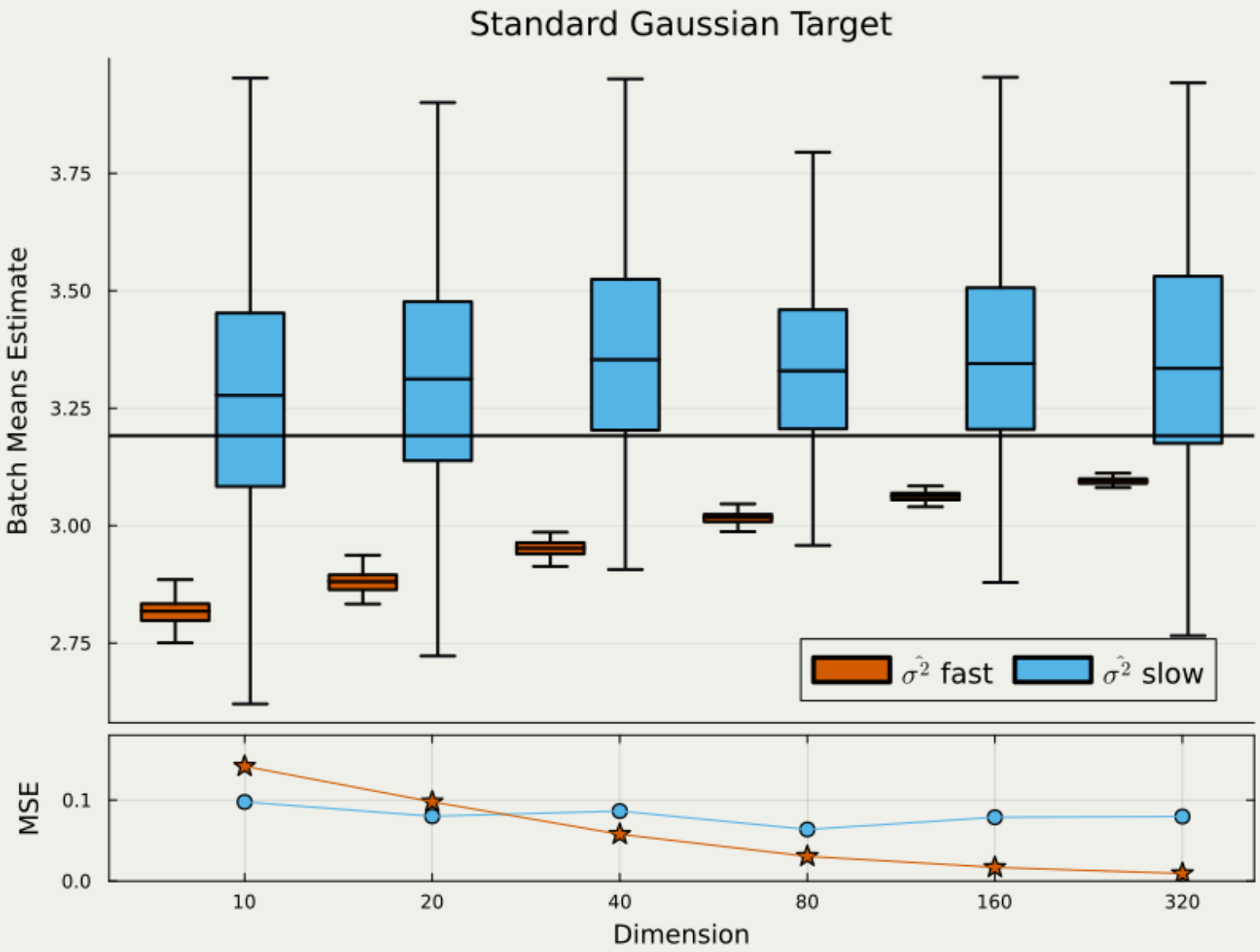


Zig-Zag Sampler
(Bierkens et al., 2019)

Bouncy Particle Sampler
(Bouchard-Côté et al., 2018)

Forward Event-Chain Monte Carlo
(Michel et al., 2020)

1.4 ESS Comparison



1.5 New Asymptotic Variance Estimator